

Aditya Arvind Desai

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EDUCATION

New York University, Center for Data Science

Sep 2024 - Dec 2026

MS, Data Science

- GPA: 3.89/4.0

- **Coursework:** Machine Learning, DL, NLP, Big Data (Hadoop, Hive), Hypothesis Testing, Mathematics (Algebra, Probability, Statistics)

Shivaji University, Rajarambapu Institute of Technology

Jul 2019 - Jul 2023

B. Tech., Electrical Engineering, Computer Science Engineering

- GPA: 3.93/4.0

EXPERIENCE

Gyori Lab, Northeastern University

Aug 2025 - Dec 2025

Machine Learning Researcher

Boston, MA, USA

- Built a hybrid entity alignment system using SeMRA priors, agent enrichment, and embeddings finetuning, improving mapping accuracy by 18% across 4 ontologies: Engineered Spark + SQL pipelines to process and evaluate 5M+ alignment candidates, reducing latency and enabling reproducible large-scale scoring, achieving 17% lift in F1 scores on OAEI benchmarks.

- Built calibration, uplift, and variance-reduction evaluations, improving sensitivity by 14%; presented findings in seminars.

Verusen AI

May 2025 - Sep 2025

AI Engineer (Intern)

Atlanta, Georgia, USA

- Architected and deployed Verusen Material AI (VMAI), a production-grade generative AI multi-agent system powered by large language models (GPT-4, Claude) and a custom-built UNSPSC model for autonomous inventory optimization and supply-chain decision-making, integrating agent-based workflows with planning, tool usage, and multi-step reasoning through LangGraph and LangChain.

- Led development of a hierarchical UNSPSC classification engine spanning 57 segments, 566 families, and 7,997 classes by combining graph modeling with Neo4J, vector databases and embeddings, and transformer fine-tuning with PyTorch.

- Built agentic workflows through LangGraph state graphs for planning and tool/API calling, incorporating agent memory with MongoDB and vector databases and guardrails including HITL, reflection, citation, and response verification for reliable multi-turn decisions.

- Built production APIs with FastAPI and Flask that handled 10K+ daily requests at less than 200ms p95, alongside data pipelines using Azure Databricks and Spark to process 5M+ records. Integrated Azure AI Search and MongoDB, reducing inference latency by 16%.

- Optimized distributed GPU/CPU clusters for LLM fine-tuning and inference, implementing CI/CD pipelines (Jenkins, GitLab) with automated testing and monitoring to ensure 99.5% uptime in production.

- Applied causal inference, A/B testing, and operations-research optimization to quantify ROI. Collaborated with cross-functional teams and presented insights to C-suite stakeholders, directly influencing strategic decisions.

Storyline Health

Feb 2025 - May 2025

Data Scientist (Intern)

Salt Lake City, Utah, USA

- Designed Storyline Behavioral AI, a healthcare informatics multi-agent system for automated patient monitoring, real-time alerting, and assistance.

- Built a retrieval-augmented generation (RAG) pipeline using LangGraph, OpenAI, AWS, OpenSearch, and RAGAS, and developed six HIPAA-aware generative AI evaluation metrics.

- Ran A/B and causal inference experiments to reduce LLM hallucinations; performed EDA to optimize alert accuracy.

Seamless Digital Automations

Jun 2023 - May 2024

Software Engineer

Mumbai, MH, India

- Engineered data acquisition and transformation pipelines for 5,000+ financial funds across 600 sources; applied statistical validation, anomaly detection, and feature engineering to create analysis-ready datasets for forecasting and risk modeling.

- Developed SeamOCR, an OCR + NLP workflow for automated document interpretation; using Python, Flask, React.js, OpenCV, Tesseract, and spaCy; implemented modular parsing components & confidence-scoring logic reducing processing errors by 91%.

PROJECTS

Rec-Sense: Social Sentiment-Aware Recommendation Engine [Demo/GitHub](#)

- Built a hybrid recommendation engine combining content-based filtering with real-time social sentiment from YouTube, Twitter, and IMDb using a fine-tuned transformer model that achieved 91% F1. Designed the SASS algorithm with cosine similarity and temporal weighting to generate personalized top-10 recommendations while addressing the cold-start problem.

- Developed production APIs (Flask) serving real-time recommendations. Conducted A/B testing, Bayesian modeling, and uplift analysis to optimize LTV/ROI metrics, improving engagement by 34% through sentiment drift modeling.

BertMap: Hybrid Ontology Alignment & Enrichment Pipeline [GitHub](#)

- Engineered a hybrid biomedical ontology alignment system integrating SeMRA priors, agentic RAG enrichment, and BERTMap/SapBERT fine-tuning, improving cross-ontology mapping recall and semantic consistency across HP, MP, GO, and FMA.

- Designed a large-scale Spark + SQL evaluation pipeline processing 5M+ candidate mappings with reproducible scoring, boosting F1 by 17% on OAEI-style benchmarks while reducing alignment latency through optimized distributed compute.

- Built agent-based enrichment modules inspired by CurateGPT to expand definitions, synthesize synonyms, and normalize lexical variants, raising downstream embedding quality. Implemented a robust embedding fine-tuning workflow using contrastive training over SeMRA-labeled pairs, improving mapping accuracy and tightening embedding-space clustering.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C, C++, Java, SQL, FastAPI, Flask, PyTorch, Pandas, NumPy, Excel, Tableau, Power BI, Looker

- **Agentic AI & LLMs:** Generative AI, Artificial General Intelligence, Generative AI Model Development, Large Language Models, LangGraph, LangChain, LangSmith, LlamaIndex, OpenAI Agents SDK/Swarm, GPT-4, Claude, LLaMA, Google Gemini, RAG, semantic search, tool/function calling, agent memory, guardrails, HITL, reflection, grounding/citation, response verification, hallucination mitigation, prompt engineering, fine-tuning, MLOps, Vector Databases, Vector Embeddings, Deep Learning

- **Data Science, Cloud & DevOps:** Healthcare Informatics, Cloud Services - AWS, Azure AI Search, Chroma, OpenSearch, Elasticsearch, Neo4J, MongoDB, PostgreSQL, Spark, Databricks, ETL, Data Pipelines, data modeling, analytics, AWS Bedrock, SageMaker, Vertex AI, Azure, GCP, Docker, Kubernetes, CI/CD, Terraform, deployment, API development, Cursor, Copilot, A/B Testing, Experimentation, EDA, HIPAA, Hyperparameter tuning